Attracting Cycles and Critical Invariant Curves in Practical Adam-SHANG

Abstract

For the practical scalar Adam-SHANG recurrence we prove a nonminimizing quadratic two-cycle, its exact fixed-curvature stability interval, nonlinear attraction at the oscillatory threshold, and a supercritical Neimark–Sacker branch consisting of two exchanged attracting curves. A fully specified quartic polynomial gives a direct critical Lyapunov certificate. Exact interval arithmetic also certifies convergence from a zero-momentum initialization. The results concern the practical parameterization, not a contradiction of the source's separately conditional convergence theorem.

Keywords: Adam–SHANG; attracting two-cycles; Neimark–Sacker bifurcation; critical invariant curves; Lyapunov stability; validated initialization

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1 Practical recurrence and initialization

Take deterministic gradients, dimension one, and the practical Algorithm 1 of Yu, Chen and Feng [1] with $\lambda = \beta = \gamma = 1$. On $f(x) = ax^2$, $a > 0$, its full state is $Z = (x, y, p, r)$, where r is the lagged metric. Write

$$\alpha = \sqrt{p+\varepsilon}, \qquad u = \frac{x + \alpha y - 2a\alpha x/(r+\varepsilon)}{1+\alpha},$$
$$v = y - \frac{2a\alpha u}{p+\varepsilon}, \qquad w = \frac{p + \alpha(2au)^2/(p+\varepsilon)}{1+\alpha}, \qquad T(Z) = (u, v, w, p). \tag{1}$$

The symbol u denotes the new primal iterate. On $\mathcal{D} = \{p > 0, r > 0\}$, this map is smooth and preserves $\mathcal{D}$. The source permits initial $r_0 = p_0 > 0$ with freely specified x_0, y_0. The conventional zero-momentum submanifold is $y_0 = x_0$, $r_0 = p_0 > 0$. Let $S = \mathrm{diag}(-1, -1, 1, 1)$. Direct inspection gives $TS = ST$.

The initial cycle and initialization results below are included as background needed to state subsequent stability and critical results self-containedly. None of the arguments freezes the metric or omits its lag.

2 An attracting quadratic cycle family

Theorem 1. *For every fixed $a > 0$, set $\varepsilon = 3a^2/4$. The two states*

$$Z_+ = (a/4, -a/4, a^2/4, a^2/4), \qquad Z_- = SZ_+$$

form a locally asymptotically attracting two-cycle of (1) *in the full four-dimensional domain. Its objective value is* $a^3/16 > 0$*, whereas the minimum is zero. Its basin contains a nonempty relatively open subset of* $\{r = p > 0\}$*.*

Proof. At Z_+, $\alpha = a$ and both metric denominators equal a^2. Substitution gives $u = -a/4$, $v = a/4$, $w = a^2/4$. Symmetry proves the other step. Scale $x = aX$, $y = aY$, $p = a^2P$, $r = a^2R$, and write $s = \sqrt{P + 3/4}$. The conjugate map is

$$u = \frac{X + asY - 2sX/(R + 3/4)}{1 + as}, \quad v = Y - 2u/s, \quad w = \frac{P + 4au^2/s}{1 + as}, \quad F = (u, v, w, P).$$

At $q_+ = (1/4, -1/4, 1/4, 1/4)$ its full Jacobian is

$$J = \begin{pmatrix} -\frac{1}{a+1} & \frac{a}{a+1} & -\frac{1}{4(a+1)} & \frac{1}{2(a+1)} \\ \frac{2}{a+1} & \frac{1-a}{a+1} & \frac{1-a}{4(a+1)} & -\frac{1}{a+1} \\ \frac{2a}{(a+1)^2} & -\frac{2a^2}{(a+1)^2} & \frac{-a^2+5a+4}{4(a+1)^2} & -\frac{a}{(a+1)^2} \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Thus $D(F^2)(q_+) = (SJ)^2$. Determinant expansion gives

$$\begin{aligned} \det(zI - SJ) &= \frac{zC_a(z)}{4(a+1)^2}, \\ C_a(z) &= 4(a+1)^2z^3 - (3a^2 + 9a + 4)z^2 \\ &\quad + (a^2 + 2a - 4)z + 3a + 4. \end{aligned}$$

The Cayley substitution $z = (1 + h)/(1 - h)$ gives

$$\begin{aligned} (1-h)^3 C_a((1+h)/(1-h)) &= 2A_a(h), \\ A_a(h) &= 4a(a+2)h^3 + (7a^2 + 20a + 16)h^2 + (4a^2 + 2a)h + a(a+2). \end{aligned}$$

All coefficients b_3, b_2, b_1, b_0 are positive and

$$b_2b_1 - b_3b_0 = 2a(12a^3 + 39a^2 + 44a + 16) > 0.$$

The cubic Routh–Hurwitz criterion puts all roots of A_a in the open left half-plane. There is no lost root since $C_a(-1) = -8a(a + 2) \neq 0$. The inverse Cayley map sends that half-plane to the unit disk, so SJ and $(SJ)^2$ are strictly Schur stable, including their zero eigenvalue.

For a smooth map with derivative of spectral radius less than one, an equivalent norm makes the derivative norm less than one. By continuity a small convex ball about q_+ is mapped into itself contractively by F^2. Even iterates converge to q_+; continuity gives odd convergence to Sq_+. Matching neighborhoods yield Lyapunov stability of the cycle. Undoing the scaling gives open attraction neighborhoods whose intersections with $r = p > 0$ are nonempty and relatively open. ∎

This family permits small ε only while changing curvature a. It makes no uniform-in-a or global-basin assertion.

3 Exact fixed-curvature stability boundary

From now on let $a = 1$ and $0 < e = \varepsilon < 1$. Write $q = \sqrt{1-e}$.

Theorem 2. *The states $Z_+ = (q/2, -q/2, q^2, q^2)$ and $Z_- = SZ_+$ form a two-cycle. Its exact strict Schur stability interval is*

$$e_* < e < 1, \qquad e_* = (8 - 2\sqrt{10})/3.$$

For $0 < e < e_*$, two return multipliers lie outside the unit disk and the cycle is unstable. At $e = e_*$ the return multipliers are

$$0, \quad (7 - 2\sqrt{10})/2, \quad (1 + i\sqrt{15})/4, \quad (1 - i\sqrt{15})/4.$$

Proof. On the cycle $p + e = r + e = 1$, so $\alpha = 1$ and substitution proves the two steps. In original coordinates the derivative at Z_+ is

$$J = \begin{pmatrix} -1/2 & 1/2 & -q/4 & q/2 \\ 1 & 0 & 0 & -q \\ q & -q & 1/2 & -q^2 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Again the return derivative is $(SJ)^2$. Its characteristic polynomials are

$$\begin{aligned} \chi_{SJ}(z) &= zC_e(z)/4, \quad C_e(z) = 4z^3 - 4z^2 + (2 - 3e)z + 4 - 3e, \\ \chi_{(SJ)^2}(t) &= tD_e(t)/16, \\ D_e(t) &= 16t^3 - 24et^2 + (9e^2 - 36e + 36)t - 9e^2 + 24e - 16. \end{aligned}$$

The Cayley transform of C_e is

$$6h^3 + (26 - 6e)h^2 + (12e - 6)h + 6(1 - e),$$

with no lost root since $C_e(-1) = -6$. Its Hurwitz determinant is

$$(26 - 6e)(12e - 6) - 36(1 - e) = -24(3e^2 - 16e + 8).$$

For $0 < e < 1$, strict Hurwitz stability is equivalent to $e > 1/2$ and $3e^2 - 16e + 8 < 0$, hence $e > e_*$. Below e_* the Routh first-column signs are $+, +, -, +$, giving exactly two right-half-plane roots and therefore two multipliers outside the disk. This gives nonlinear instability by the differentiable-map instability theorem. At equality, setting $\rho = 1 - \sqrt{10}/2$ gives

$$C_{e_*}(z) = 4(z - \rho)(z^2 - \sqrt{10}z/2 + 1).$$

Squaring the roots gives the stated return multipliers. Strict stability implies local nonlinear attraction as in Theorem 1, but equality needs the next sections. ∎

For fixed curvature, truly small e is therefore unstable. As $e \uparrow 1$, the cycle coalesces at the zero-metric boundary of $\mathcal{D}$; the limiting characteristic polynomial is $t(t-1)(t-1/4)^2$. We do not classify that endpoint.

4 Critical nonlinear stability and exchanged invariant curves

Theorem 3. *At $e = e_*$ the cycle is locally asymptotically stable. For $e < e_*$ sufficiently close to e_*, a supercritical Neimark–Sacker bifurcation produces two disjoint small attracting closed curves $\mathcal{C}_\pm$, one about each cycle phase. The original map exchanges them, and its square preserves each. No claim is made that all dynamics on these curves are quasiperiodic.*

Proof. Use $G_e = ST_e$, so $G_e^2 = T_e^2$. For $k = 1 - e$, scale with $D = \mathrm{diag}(\sqrt{k}, \sqrt{k}, k, k)$; the fixed phase is $b = (1/2, -1/2, 1, 1)$. In scaled coordinates,

$$\alpha = \sqrt{1 + k(P-1)}, \quad U = \frac{X + \alpha Y - 2\alpha X/(1 + k(R-1))}{1+\alpha},$$
$$\widetilde{G}_e(X, Y, P, R) = (-U, -Y + 2U/\alpha, (P + 4U^2/\alpha)/(1+\alpha), P).$$

Center at b and write $F(h) = Ah + B(h,h)/2 + C(h,h,h)/6 + O(\|h\|^4)$ at e_*. Thus B, C are raw derivatives, not polynomial coefficients. Here $k = (2\sqrt{10} - 5)/3$ and

$$A = \begin{pmatrix} 1/2 & -1/2 & k/4 & -k/2 \\ -1 & 0 & 0 & k \\ 1 & -1 & 1/2 & -k \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

Its eigenvalues are $\mu = (\sqrt{10} + i\sqrt{6})/4$, $\bar{\mu}$, 0, and $\rho = 1 - \sqrt{10}/2$. Choose

$$q_0 = \begin{pmatrix} -2/3 + 7\sqrt{10}/24 + i\sqrt{6}/8 \\ 3/4 - \sqrt{10}/4 + i(\sqrt{6} - \sqrt{15})/4 \\ \mu \\ 1 \end{pmatrix},$$

and the unique complex row ℓ satisfying $\ell A = \mu\ell$, $\ell q_0 = 1$. Then $\ell \bar{q}_0 = 0$. Define

$$h_{11} = (I - A)^{-1} B(q_0, \bar{q}_0), \qquad h_{20} = (\mu^2 I - A)^{-1} B(q_0, q_0),$$
$$c = \tfrac{1}{2}\ell\{C(q_0, q_0, \bar{q}_0) + 2B(q_0, h_{11}) + B(\bar{q}_0, h_{20})\}.$$

The inverses exist by the displayed eigenvalues. Exact arithmetic in $\mathbb{Q}(\sqrt{10}, i\sqrt{6})$ gives

$$c = \frac{951625}{5067648} - \frac{5081\sqrt{10}}{140768} + i\left(\frac{9447395\sqrt{6}}{3800736} - \frac{24833969\sqrt{15}}{15202944}\right),$$
$$d = \mathrm{Re}(\bar{\mu}c) = \frac{9218750 - 2985293\sqrt{10}}{2533824} < 0.$$

The strict sign is certified by $10 \cdot 2985293^2 - 9218750^2 = 4134391395990 > 0$. These are the normal-form conventions of [4]. The supplied symbolic certificate constructs the cubic jet

by exact truncated multiplication and inversion and checks every derivative against direct differentiation of the untruncated map. Thus the factor $1/2$ and conjugation convention above are fixed, not inferred from decimal eigenvalues.

Implicit differentiation of $C_e(\mu(e)) = 0$ gives

$$\mu'(e_*) = \frac{3(\mu+1)}{12\mu^2 - 8\mu + 2 - 3e_*}, \qquad \left.\frac{d|\mu(e)|}{de}\right|_{e_*} = -\frac{45+21\sqrt{10}}{212} < 0.$$

Exact algebra verifies $\mu^j \neq 1$ and $\mu^{2j} \neq 1$ for $j = 1, 2, 3, 4$. The stable eigenvalues remain strictly inside the disk. The center-manifold and Neimark–Sacker hypotheses therefore hold. The zero stable eigenvalue causes no obstruction: the smooth-map center-manifold theorem [5] needs no inverse on the stable block, while the two-dimensional center restriction is locally invertible.

On the center manifold the normal form is $w' = \mu w + cw|w|^2 + O(|w|^4)$, so the critical radial decrement is $2d|w|^4 + O(|w|^5) < 0$. Full-space attraction at equality follows either from the polynomial certificate in the next section or by flattening the center manifold: transverse displacement contracts as $\|\eta'\| \le a\|\eta\|$, $a < 1$, while the error in the squared-radius decrement is bounded by $C\|\eta\|$. The function $|w|^2 + M\|\eta\|$, with $M > C/(1-a)$, strictly decreases on a sufficiently small neighborhood, and summing its decrement proves convergence and stability.

Put $\delta = e - e_*$. The radial unfolding parameter is $\sigma = |\mu(e)| - 1 = -(45 + 21\sqrt{10})\delta/212 + O(\delta^2)$. Since $d < 0$, the attracting invariant-curve branch is on $\sigma > 0$, that is $\delta < 0$, with squared radius $-\sigma/d + o(|\sigma|)$. Stable transverse contraction extends attraction to four dimensions. Finally if $\mathcal{C}_+$ is the local G-invariant curve and $\mathcal{C}_- = S\mathcal{C}_+$, then $T\mathcal{C}_+ = SG\mathcal{C}_+ = \mathcal{C}_-$ and $T\mathcal{C}_- = \mathcal{C}_+$. They are disjoint at sufficiently small amplitude. This proves the stated geometry for the original algorithm. Also the twice-composed cubic coefficient is $2\mu c$, whose radial coefficient is $2d < 0$, consistently with $G^2 = T^2$. ∎

5 A fully specified quartic Lyapunov polynomial

The following finite formula specifies every coefficient without a hidden choice. It is accompanied by machine-readable expanded coefficients and an independent original-map checker. Use the centered F above and let

$$v_0 = (k, 0, 0, 1)^T, \quad v_\rho = (-1/6 - \sqrt{10}/12, -3/2, \rho, 1)^T, \quad E = [q_0, \bar{q}_0, v_0, v_\rho].$$

For $y = (z, \bar{z}, s, t) = E^{-1}h$ and $\Lambda = \mathrm{diag}(\mu, \bar{\mu}, 0, \rho)$, write $f(y) = E^{-1}F(Ey) = \Lambda y + f_2(y) + f_3(y) + O(\|y\|^4)$, where f_j is its homogeneous Taylor part. Treat $z, \bar{z}$ as formal independent variables during polynomial operations. Let

$$Q(y) = z\bar{z} + s^2 + t^2, \qquad \lambda^a = \mu^{a_1}\bar{\mu}^{a_2}0^{a_3}\rho^{a_4}, \quad 0^0 = 1.$$

Define the finite polynomials

$$\begin{aligned}
R_3 &= DQ(\Lambda y) f_2(y),\\
V_3 &= -\sum_{|a|=3} \frac{[y^a]R_3}{\lambda^a - 1} y^a,\\
R_4 &= Q(f_2(y)) + DQ(\Lambda y) f_3(y) + DV_3(\Lambda y) f_2(y),\\
V_4 &= -\sum_{\substack{|a|=4\\ a\neq(2,2,0,0)}} \frac{[y^a]R_4}{\lambda^a - 1} y^a,\\
V(h) &= (Q + V_3 + V_4)(E^{-1}h).
\end{aligned}$$

The resonant coefficient of V_4 is fixed to zero. These expressions involve only differentiation, finite sums and explicitly given matrices.

Theorem 4. *The real degree-four polynomial V has positive-definite quadratic part and satisfies*

$$V(F(h)) - V(h) = -s^2 - (1-\rho^2)t^2 + r_4|z|^4 + O(\|h\|^5),$$

where

$$r_4 = \frac{5690524925 - 1802647064\sqrt{10}}{1812951072} < 0.$$

It is a strict local Lyapunov function and gives a phasewise one-step polynomial certificate for the critical original cycle.

Proof. Every monomial involving the zero direction has $\lambda^a = 0$, and every monomial involving the other stable direction has $|\lambda^a| < 1$. Center-only cubic exponent differences are $\pm 1, \pm 3$, so none resonate. At degree four the possible differences are $0, \pm 2, \pm 4$; only $(2,2,0,0)$ resonates. Hence all denominators in the definitions are nonzero. The scalar pullback operator on y^a multiplies it by $\lambda^a - 1$. Consequently V_3 cancels every cubic residual and V_4 every nonresonant quartic residual. Exact expansion of the sole remaining coefficient gives $[z^2\bar{z}^2]R_4 = r_4$. Its strict sign follows from

$$10 \cdot 1802647064^2 - 5690524925^2 = 113290451431945335 > 0.$$

Conjugation exchanges the first two coordinates and fixes the stable ones; the real map, Q, and unique prescribed homological solves respect this involution. Thus V is real for real h. Its quadratic part is $|z|^2 + s^2 + t^2 > 0$ away from zero.

For precision about the finite algebraic certificate, V_3 has 19 nonzero terms and V_4 has 34. Together with Q there are 56 eigen-coordinate terms and 65 expanded real-state terms. The accompanying exact coefficient file and direct checker verify every coefficient through degree four of $V(F(h)) - V(h) + s^2 + (1-\rho^2)t^2 - r_4(z\bar{z})^2$ is zero, differentiating the original radical map rather than importing a center normal form. In these unflattened

coordinates r_4 need not equal $2d$; stable quadratic decrement contributes at quartic order along a curved center manifold.

Analyticity bounds the remainder by Cn^5, where $n^2 = |z|^2 + s^2 + t^2$ and $u^2 = s^2 + t^2$. For $n \le \delta \le 1$,

$$n^5 \le 2\delta|z|^4 + 2\delta^3 u^2.$$

Choose δ so that $2C\delta \le -r_4/2$ and $2C\delta^3 \le \min(1, 1-\rho^2)/2$. Then

$$V(F(h)) - V(h) \le (r_4/2)|z|^4 - \tfrac{1}{2}\min(1, 1-\rho^2)(s^2+t^2).$$

After further shrinking, $n^2/2 \le V(h) \le 3n^2/2$. A small compact sublevel set remains in the coordinate ball and is invariant by continuity and decrease. Summing the decrement proves $z, s, t \to 0$; the quadratic bounds prove stability.

For physical phase neighborhoods define

$$W_+(Z) = V(D^{-1}Z - b), \qquad W_-(Z) = V(D^{-1}SZ - b).$$

Since S commutes with D and T, $W_-(TZ) - W_+(Z) = V(F(h)) - V(h)$ near Z_+, and symmetrically at Z_-. Matching sublevels are exchanged and strictly decrease away from the cycle. These are phasewise polynomials, not a single asserted global polynomial. No quantitative basin radius is claimed. ∎

6 Certified zero-momentum initialization

Theorem 5. *For $a = 1$, $\varepsilon = 3/4$, the exact initial state $Z_0 = (1/4, 1/4, 1/4, 1/4)$ converges to the two-cycle $Q_+ = (1/4, -1/4, 1/4, 1/4)$, $Q_- = SQ_+$, with even iterates tending to Q_+. A relatively open neighborhood of Z_0 in $\{y = x, r = p > 0\}$ has the same limit.*

Proof. Set $G = ST$ and $B = \{\|Z - Q_+\|_\infty \le 2^{-60}\}$. The exact integer interval certificate `initialization.py`, using 2048 binary fractional bits, proves all intermediate G iterates through 20 steps are defined on B and

$$\sup_B \|D(G^{20})\|_\infty < 1/4.$$

The upward rounded row-sum bounds are respectively 0.182442974107, 0.194975439374, 0.241166792098, 0.185380815812. Direct enclosure of the original orbit from its exact rational state proves

$$\|T^{400}(Z_0) - Q_+\|_\infty < 2^{-68} < 2^{-60}.$$

No orbit enclosure is reset to a floating-point center. Since $G^{20}(Q_+) = Q_+$ and B is convex, the derivative bound makes $G^{20} = T^{20}$ a contraction of B into itself. Thus $T^{400+20n}(Z_0) \to Q_+$. Continuity of each T^j, $0 \le j < 20$, supplies all residue classes, proving even/odd convergence. Finally the 400-step image lies strictly inside B, so continuity gives an open neighborhood of Z_0 whose images also lie there; intersecting with the zero-momentum initialization manifold proves the final statement. ∎

7 Source-specific interpretation

The practical Algorithm 1 of [1] is distinct from its equation (5) and Theorem 2.1, whose convergence statement imposes bounded auxiliary trajectories, coupled parameters and an admissible step. Fixed positive stabilization and the practical parameterization here are not that coupled schedule. The verified nonminimizing cycle is therefore not a counterexample to that conditional theorem. The deterministic companion [2] likewise separates its exact conditional schedule from stabilized practical methods. No false published theorem is alleged on this basis.

Quadratic Adam cycles are known in closely related dynamics [3]. What is specific here is the full lagged-preconditioner SHANG recurrence, its trace-dependent mixing, exact fixed-curvature boundary, critical polynomial certificate and initialized basin. Shared constant-metric alternating behavior does not establish a conjugacy to ordinary Adam away from the cycle. No worldwide novelty, global attraction, all-initializations result, neural-training conclusion, or statement about a different software implementation is made. In particular the critical theorem and initialization theorem concern different values of ε and must not be combined into a claim of certified conventional initialization at e_*.

A Reproducibility and exact arithmetic

The accompanying certificate directory contains subdirectories for the family, stability interval, normal form and polynomial Lyapunov function, plus the root-level initialization script. Symbolic scripts require Python 3 and SymPy; the initialization script uses only the standard library. Run the scripts in their own directories. File `README.txt` gives the precise commands.

The symbolic normal-form script constructs the cubic jet from $\sqrt{1+kh_P} = 1 + kh_P/2 - k^2h_P^2/8 + k^3h_P^3/16 + O(\|h\|^4)$ and finite truncated multiplication/inversion over $\mathbb{Q}(\sqrt{10}, i\sqrt{6})$. A monomial coefficient at multi-index a corresponds to raw derivative $a!$ times that coefficient. The direct differentiation checker verifies all 136 symmetric derivatives of orders one through three. The Lyapunov checker reads only the proposed coefficient data, defines the untruncated original map, and verifies the residual in original real coordinates; derivatives only through order three suffice since V starts at degree two.

For the initialization certificate an interval $[L, U]$ means $[L/2^{2048}, U/2^{2048}]$. Addition and negation are exact. Products use all endpoint products with outward integer quotient rounding. Reciprocals check nonzero sign and reverse endpoints. Square roots use lower integer floor and upper integer ceiling of the exact integer square root of the appropriately scaled endpoint. Forward differentiation propagates four interval derivatives with the same enclosing arithmetic. Induction therefore encloses every Jacobian over the entire starting box; row sums bound the induced infinity norm. These arithmetic arguments and

the contraction proof establish an exact finite certificate, not a floating-point trajectory inference.

References

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